F28GE109-CB

400A
4 POLE

G
3~

F28-GE-109
415V
250kVA

F28-SR-002
415V ULTRA FINE GRINDING SWITCHROOM

SWR EARTH BAR

FEEDER FROM F28-SW-009
2000-F28-SLD-EL-10201
TO F28-SW-009
2000-F28-SLD-EL-10201

F28-SW-109
ULTRA FINE GRINDING 415V ESSENTIAL SWITCHBOARD

EARTH BAR

NEUTRAL BAR
NOTE 1

AFD

125A
160A

SD
NOTE 3

2A
63A

2A
63A

PFR

2A
63A

00F-05

MANUAL
MCCB CHANGEOVER
SWITCH 3 POLE 400A

415V 3NE~ 50Hz 400A 65kA for 1sec

INC

01F-17
250A
2A
32A
90A
250A
250A

1F-01
32A
2A
32A
70A
32A
32A

01F-09
250A
2A
32A
90A

01F-07
63A
2A
32A
70A

01F-05
63A
2A
32A
70A

1F-05
250A
2A
32A
90A

TEMPORARY
GENERATOR

F28-SR-001
11kV ULTRA FINE
GRINDING
SWITCHROOM

BC
3.6kW

F28-BC-001
ULTRA FINE GRINDING
11kV SWITCHROOM
48V CHARGER
2000-F28-TLD-EL-10051

69kW

F28-DB-001
ULTRA FINE GRINDING
11kV / 415V SWITCHROOM
L&SP DB
2000-F28-SCM-EL-10205

ISL
32A

F28UPS002-ISL
ULTRA FINE GRINDING
415V/240VAC UPS
BYPASS SWITCH
2000-F28-SCM-EL-10206

10kW

F28-DB-201
ULTRA FINE GRINDING
415V SWITCHROOM
240VAC UPS DB
2000-F28-TLD-EL-10054

UPS
10kW

F28-UPS-002
ULTRA FINE GRINDING
415V SWITCHROOM
240VAC UPS
2000-F28-TLD-EL-10054

69kW

F28-DB-301
ULTRA FINE GRINDING AREA
L&SP DB
2000-F28-SCM-EL-10207

63A
SPARE FEEDER

F28CR430-JB02
UFG AREA
CRANE

JB

CR
63A

F28-CR-430
UFG AREA
CRANE
2000-F28-SCM-EL-10209

JB

F72DB302-JB
REAGENTS MIXING &
STORAGE AREA L&SP DB 2
JUNCTION BOX

69kW

F72-DB-302
REAGENTS MIXING & STORAGE AREA
L&SP DB 2
2000-F72-SCM-EL-10406

NOTES

1. EARTH-NEUTRAL LINK INSTALLED IN UPSTREAM F28-SW-009 INCOMER TIER. SIGNAGE SHALL BE PLACED ON F28-SW-109 SWITCHBOARD AND ON GENERATOR READING "M.E.N. LINK LOCATED IN F28-SW-109 INCOMER TIER"
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
4. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
5. CIRCUIT BREAKER TO BE SUBSTITUTED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.

LEGEND

PFR	PHASE FAILURE RELAY
I >	EARTH LEAKAGE RELAY
xxxA / xxxA	ST TRIP SET POINT / CB SIZE
AFD	ARC FLASH DETECTION
ECR	EARTH CONTINUITY RELAY

DRG No.	REFERENCES
2000-F28-SCM-EL-10204	F28-SW-009 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F28-SCM-EL-10203	F28-SW-009 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F28-SCM-EL-10220	F28-SW-109 415V ESSENTIAL SWITCHBOARD INCOMER SCHEMATIC DIAGRAM
2000-F28-DRG-EL-10001	F28-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F28-BLK-IC-10001	F28-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F28-SLD-EL-10201	F28-SW-009 415V MCC SINGLE LINE DIAGRAM SHEET 1
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	02OCT26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
1	15SEP25	FAT REDLINES INCORPORATED					
0	29AUG24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ULTRA FINE GRINDING 2/3
F28-SW-109 415V ESSENTIAL SWITCHBOARD
SINGLE LINE DIAGRAM

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F28-SLD-EL-10211	2

O:\CAD PROJECTS\23S NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F28 - ULTRAFINE GRINDING 2 & 3\2000-F28-SLD-EL-10211.DWG 29SEP26 7:42AM ROBINLAW